# Mitigating Global Cereal Crop Loss: Characterization and Inhibition of a Cdc14 Phosphatase in *Claviceps purpurea*

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Running title: Mitigating Cereal Crop Loss via *CpCdc14* Phosphatase Inhibition

## Abstract

Globally, food insecurity and undernourishment are exacerbated by crop pathogens, pests, and abiotic stressors. *Claviceps purpurea* (Cp) is a pathogenic fungus that causes ergot infection in rye, wheat, barley, triticale, and other cereal crops. To date, strategies to mitigate Cp infection have been largely unsuccessful. A gene coding for the dual-specificity phosphatase Cdc14 has been shown to be critical for the related fungus *F. graminearum* to infect wheat; absence of the *CDC14* gene prevents fungal infection in plants. We hypothesize that inhibition of Cdc14 would have the same effect in *C. purpurea.* Importantly, *CDC14* is conserved across ascomycetes but is largely absent in angiosperms. To test our hypothesis, we overexpressed and purified CpCdc14 in a pET-15b *E. coli* recombinant plasmid. We demonstrated that the purified enzyme had phosphatase activity using pNPP as a model substrate. We compared the substrate specificity of CpCdc14 to the specificity observed in other fungal pathogens as well as in the budding yeast *S. cerevisiae*. In addition, we tested a panel of small molecules for their ability to inhibit the enzyme. Moreover, given the high degree of sequence similarity between ScCdc14 and CpCdc14, we built a homology model of CpCdc14 using ScCdc14 as a template. Both *in vitro* and *in silico* experiments identified aurintricarboxylic acid (Inhibitor I1) as a lead compound for future optimization. Taken together, our results suggest that inhibition of Cdc14 is indeed a viable strategy for development of a fungicide with broad-spectrum application.

*Keywords: enzyme inhibitor, dual-specificity phosphoprotein phosphatase, molecular modeling, enzyme kinetics, protein-ligand docking, global food insecurity, fungal pathogens*

## Introduction

*Broad Problem and Fungal Introduction*

Globally, 828 million individuals in 2021 faced hunger, which is an increase of roughly 150 million compared to 2019 (1). Undernourishment can result from food insecurity and is projected to be exacerbated over the 21st century, due to global trade policies, unprecedented climate change, crop pathogens, pests, and other abiotic stressors (2). One notable crop pathogen to discuss is the fungal species *Claviceps purpurea,* which produces toxic ergot alkaloids, causing ergot in rye, barley, millet, triticale, wheat, and sorghum (3). These cereal crops are vital for the international agricultural economy, as the consumption of cereals is projected to increase from 2.7 billion tons in 2021 to 3.0 billion tons in 2030 (4). Therefore, *C. purpurea* can magnify undernourishment and cause economic detriment, prompting the need to investigate potential fungal inhibition.

*Mechanistic Effects of Claviceps purpurea*

*Claviceps purpurea* is the most common member of the genus *Claviceps* (5). As an ergot fungus, its infection of a host crop begins in the spring when a wind-born ascospore lands on the plant's stigma (6]\). Fungal hyphae begin to invade and infect the ovary, as a sticky fluid called "honeydew" develops in the stroma (7). Flies and other small insects are attracted by this fluid, which can ultimately spread the *C. purpurea* around the plant. Within four to five weeks, a mass of mycelium called a sclerotium will replace the seed, indicating full host crop colonization (8). Once the seed is gone, there is nothing of the plant left to harvest. In addition to causing cereal crop failure, *Claviceps purpurea* can result in negative human health effects. Consumption of these compounds or ergot sclerotia can result in hallucinations, muscle spasms, vomiting, and reduced blood flow (9).

*Implications of Cdc14 Family Phosphatases*

An enzyme identified in *Claviceps purpurea* is a member of the dual-specificity phosphatase domain of cell division control protein 14 (Cdc14) family proteins; DNA sequence BLAST analyses confirmed this. Cdc14 phosphatases counteract cyclin-dependent kinases (CDKs) and other kinases at the end of mitosis and cytokinesis (10). This Cdc14 phosphatase may be a favorable target for inhibition based on promising inhibition results in similar fungal targets like *Saccharomyces cerevisiae* (Sc) (11).

One observation when Cdc14 family proteins were inhibited or removed is that numerous pathogenic fungal species exhibited impaired cytokinesis and nuclear division (12). Cdc14 phosphatases notably impact cell division and septum formation in *Fusarium graminearum,* a pathogenic fungal species (13). Furthermore, the deletion of a dual-specificity Cdc14 phosphatase in *Beauveria bassiana*, an entomopathogenic fungal species, led to defective cytokinesis and reduced overall virulence (14). Thus, there is potential in Cdc14 phosphatase inhibition affecting the ability of *Claviceps purpurea* to colonize a host cereal crop.

*Research Objective*

The primary objective is to explore the biochemical properties, enzyme kinetic parameters, and potential for inhibition of a Cdc14 phosphatase in *C. purpurea* (CpCdc14). First, the CpCdc14 will be expressed through a pET-15b recombinant plasmid and then purified using

nickel-column affinity chromatography. Purified aliquots will then be used in SDS-PAGE analysis, pNPP assays, substrate specificity assays, and inhibitor compound assays. Overall, this research can provide a novel insight in the inhibition of CpCdc14, providing a pathway for further research on a fungicide.

## Methods

*Sequence and Homology Structure Model of CpCdc14*

Our Cdc14 analog was uploaded into the Molecular Operating Environment (MOE) software, visualizing the protein in three dimensions. A PDB search was launched to determine proteins with similar structures and sequences to our analog. From this search, tyrosine protein phosphatase Cdc-14 (6G86.B) was chosen as the template, as it had the lowest non-zero HMMER E-value. The amino acid chains for the CDC14 analog were superimposed with this template protein-tyrosine phosphatase (PTP), removing any extreme outlier amino acid chains and gaps/segments that had no alignment whatsoever so that sites of interest could be analyzed. A homology model was then created for the protein colored based on root-mean-square deviation (RMSD), with the green sections being areas of high similarity, warmer red colors as areas of differences, and white as areas of absences. Lastly, a Ramachandran Plot was created, recording any unfavorable Phi-Psi, bond angles, bond lengths, atom clashes, and rotamer energies [See Figure 2C].

*Expression and Purification of CpCdc14*

A pET-15b plasmid containing the CpCdc14 gene with an N-terminal 6x-his tag and an ampicillin resistant gene was successfully transferred into competent *Escherichia coli* BL21 AI cells (Figure 3). Samples were incubated on ice for 30 minutes, 42°C for 1 minute, and then ice for 2 minutes. Cells supplemented with 2xYT media completed an outgrowth cycle at 37°C for 30 minutes. The cells were transferred to two Luria-Bertani (LB) agar plates with a final concentration of 100 µg/mL ampicillin, one at its current dilution and one at 1/100th dilution. Plates were incubated at 37°C overnight.

A single bacterial colony was used to inoculate a 5 mL 2xYT culture supplemented with 100 µg/mL ampicillin. The cells were incubated at 37° C shaking at 225 rpm overnight. One of the 5-mL starter cultures was poured into the 2.8-L Fernbach flask containing 1 L of 2xYT media supplemented with ampicillin at a final concentration of 100 µg/mL. The cultures were incubated at 22° C with shaking at 225 rpm. After about 6 hours, periodical monitoring of the culture's optical density was determined using a spectrophotometer at 600 nm until an optical density between 0.7-1.0 was reached. A 1-mL sample was collected, and L-arabinose was added to the remaining cultures in the Fernbach flasks to a final concentration of 0.1%. Flasks were returned to the incubator and allowed induction overnight at 22° C with shaking at 225 rpm.

When the large-scale bacterial culture overnight induction was complete, the optical density was measured again, and a 1 mL sample collected. After centrifugation of the remaining culture, pelleted cells were scraped into a single tared 50-mL conical tube, then stored in the -80° C freezer. The frozen pellet of cells was then thawed and mixed in a 50-mL conical tube with a 'lysis solution' containing 30 mL Lysis Buffer, 1.2 g lysozyme (1 mg/mL), 30 µL leupeptin (10 µM), 20 µL pepstatin (1 µM), and 4 µL of Universal nuclease, stirred thoroughly to a homogenous solution. This mixture was allowed to incubate on ice for 30 minutes with

periodical mixture. After 30 minutes, phenylmethylsulfonyl fluoride (PMSF) was added to a final concentration of 0.5 mM. This mixture was sonicated and then centrifuged for 30 minutes at 12,500 rpm. A 25 µL aliquot of this lysed solution was collected for later use.

A $Ni^{2+}$ nitrilotriacetic acid ($Ni^{2+}$-NTA) resin was used for purification of 6x His-tagged CpCdc14. A 1.0 mL sample of Nickel Buffer A (25 mM HEPES, 500 mM NaCl, 10% glycerol) and Ni-NTA beads were washed into a syringe over a 50 mL conical tube. Supernatant was applied to the column twice and then collected. Column was subsequently washed with Nickel Buffer A to achieve a final concentration of 40 mM imidazole. The protein was eluted with 250 mM elution into '250 mM elution 1' conical tube with Nickel Buffer B (25 mM HEPES, 500 mM NaCl, 250 mM imidazole, 10% glycerol), repeated twice. The 250 mM elution was transferred to a SnakeSkin dialysis tubing, placed in a storage buffer (25 mM HEPES, 300 mM NaCl, 2 mM EDTA, 40% glycerol) and was dialyzed overnight.

Dialyzed, purified protein concentration was then determined with a colorimetric Bradford assay of the absorbance at 595 nm. The protein was diluted 4x and 2x, averaged, and converted to concentration using the recently obtained standard curve. The concentration of the protein was verified using a nanodrop spectrophotometer with Beer Lambert's Law.

*SDS-PAGE Analysis of Purified Claviceps purpurea protein*

Sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS-PAGE) was run to determine if the *C. purpurea* enzyme had been purified. Any samples taken during purification were prepared with 2x SDS-gel loading buffer and DTT to a final concentration of 200 mM, then heated at 95°C for 5 minutes. 30 µL of each sample was loaded into a Mini-PROTEAN® TGX™ precast gel with 1x Tris-Glycine SDS Buffer and ran at 200V for 1 hour. The gel was stained with coomassie brilliant blue for 2 hours with gentle rocking, then submerged in 10% acetic acid destaining solution overnight. The gel was imaged with the Bio-Rad Gel Doc™ EZ System.

*pNPP Assay +/- Sodium Tungstate*

The activity of the *CpCdc14* enzyme was tested using a generic para-nitrophenyl phosphate (pNPP) assay. To determine optimal reaction time and concentration, an initial linearity assay was run in 96-well plates with enzyme reaction buffer (25 mM HEPES at pH 7.5, 1 mM EDTA, 150 mM NaCl), 40 mM pNPP, 1 mM DTT, and $ddH_2O$ to a final volume of 200 µL (Appendix A). The enzyme concentration varied in each well at 10, 3, 1, 0.3, and 0.1 µM.

The activity of the enzyme was then tested using the presence of sodium tungstate. Two solutions were created, one without inhibitor (25 mM HEPES at pH 7.5, 1 mM EDTA, 150 mM NaCl, 100 mM DTT, 40 mM pNPP, 1 µM Enzyme) and one with inhibitor (25 mM HEPES at pH 7.5, 1 mM EDTA, 150 mM NaCl, 100 mM DTT, 40 mM pNPP, 1 µM Enzyme, 50 mM sodium tungstate) to a final volume of 200 µL. The enzyme was allowed to react with the solution for 20 minutes, and was stopped by adding 100 µL of 5 M NaOH. The absorbance of each well was measured at 405 nm using the BioTek plate reader, and the specific activity of the enzyme was calculated.

*Michaelis-Menten Plot and Steady-State Kinetic Parameters*

Reaction well solutions were prepared with 1x enzyme reaction buffer, 1 mM DTT, 1 µM enzyme, and varying final concentrations of pNPP in diethanolamine (2, 4, 6, 8, 10, 20, 30, 40, 50, 60, 70, and 80 mM). Blanks were prepared in the same manner without the addition of enzymes. After 20 minutes elapsed, 100 µL of 5 M NaOH was added to stop the reaction from proceeding further and the assay was read by a UV/visible light spectrophotometer plate reader at 405 nm. Recorded absorbance values were converted into initial velocity ($v_0$) and graphed against initial substrate concentration to fit a Michaelis-Menten curve in Microsoft Excel. Values were optimized (minimum) using Solver and the Michaelis-Menten Equation.

*Substrate Specificity of CpCDC14*

The specificity of CpCdc14 to various amino acid sequences was tested with a specificity assay. A standard curve was first generated using sodium phosphate at various concentrations, promptly added with 100 µL Biomol Green to a final reaction volume of 145 µL, read at 640 nm. After the standard curve graph was obtained, reaction wells were prepared with 1x enzyme reaction buffer, 1 µM of enzyme, and 4 µM peptide chain to a final volume of 200 µL. Blanks were created in the same manner without the enzyme. After the reaction was left to run for 20 minutes at 25° C, 200 µL of BIOMOL Green reagent was added to every well to react with free phosphates, ($P_i$) that was released from the phosphatase-substrate reaction, then incubated for an additional 20 minutes at 25° C. The absorbance of each well was then measured at 605 nm using the BioTeplate reader. The reaction rate and $k_{cat}/K_M$ specificity constant were then calculated.

*In silico Inhibition of CpCdc14*

A high-throughput screen (HTS) in S. cerevisiae was run to identify candidate inhibitors of Cdc14 enzymes. The top 11 inhibitors identified from the HTS were uploaded to MOE along with the previously generated homology model to predict which compounds best bind to CpCdc14's active site. After setting the peptide ligand to an inert state so as not to impede with compound interactions, an induced fit receptor docking model was run for each inhibitor. The docking scores, or the S values, were computed, where the most negative S values represented the inhibitor's binding affinity to the homology model. The top three inhibitors of the docking simulation with the most ideal S values were tested again to ensure accuracy, and their ligand interaction maps were recorded and analyzed for future use.

*In vitro Inhibition of CpCdc14*

From the aforementioned HTS, the top ten inhibitors were identified as potential targets for *CpCdc14* inhibition and subject to an inhibition assay. A satisfactory Z-factor test was first completed, then reaction wells with the ten inhibitor solutions were created. Solutions contained 1x enzyme reaction buffer, 1 mM DTT, 50 mM pNPP, 1 µM enzyme, either 100 µM or 1 mM of inhibitor, and ddH2O to a final volume of 200 µL. After the reaction was run for 20 minutes, 100 µL of 5 M NaOH was added. Absorbances were then read and recorded at 405 nm.

*IC50 Screening*

The top two inhibitors (I1: Aurintricarboxylic Acid and G5: 5-(4-isopropylbenzylidene) -2,4,6(1H,3H,5H)-pyrimidinetrione) from the *in silico* and *in vitro* inhibition experiments were tested further to determine their IC50. Positive and negative controls were first prepared with 1x enzyme reaction buffer, 1 mM DTT, 40 mM pNPP, ddH2O, either 1 µM of enzyme or no enzyme, and a later added 5 M NaOH. Then, the inhibitor assay wells were prepared with 1x enzyme reaction buffer, 1 mM DTT, 1 µM enzyme, 40 mM pNPP substrate, and the inhibitors I1 and G5 at final concentrations of 1, 5, 10, 50, 100, 120, 140, 160, 180, and 200µM. The reaction ran for 20 minutes at 25℃ and was then stopped by adding 100 µL of 5M NaOH to each well. The absorbance of each well was measured at 405 nm using the BioTek plate reader. The assay was performed in triplicate. The data was then plotted as (absorbance-blank) versus log [inhibitor] and fitted to a sigmoidal curve in Excel using Solver and the equation below. A = maximum absorbance (no inhibition), B = estimate of the Hill coefficient (start with an estimate of 1), C = estimate of IC50, D = minimum absorbance (maximum inhibition).

$$y = D + \frac{A-D}{1+10^{(x-logC)B}}$$

*In silico Optimization of Enzyme-Inhibitor Interactions*

Using the Molecular Operating Environment (MOE) software, the *CpCdc14* homology model was used with 11 candidate inhibitor compounds to perform ligand docking. The results were then sorted by ascending S-value to indicate the greatest docking scores, such that lower values are more favorable. After choosing the molecular docking position with the lowest docking score and root mean square deviation (RMSD), while also referencing in vitro results, the inhibitor compound was loaded into a new MOE window with the ligand-peptide complex of *CpCdc14* for further optimization.

Following the guidelines of Lipinski's rule of five, with the expansion of the MW requirement to 750 Da, side chains, elements, functional groups, and further modifications were added to optimize the inhibitor molecule's binding affinity. Electrostatic maps, two-dimensional ligand interaction maps, and MOE Builder → Fragment was utilized to determine the most favorable modifications to the inhibitor structure.

## Results

### *Sequence and Homology Structure Model of CpCdc14*

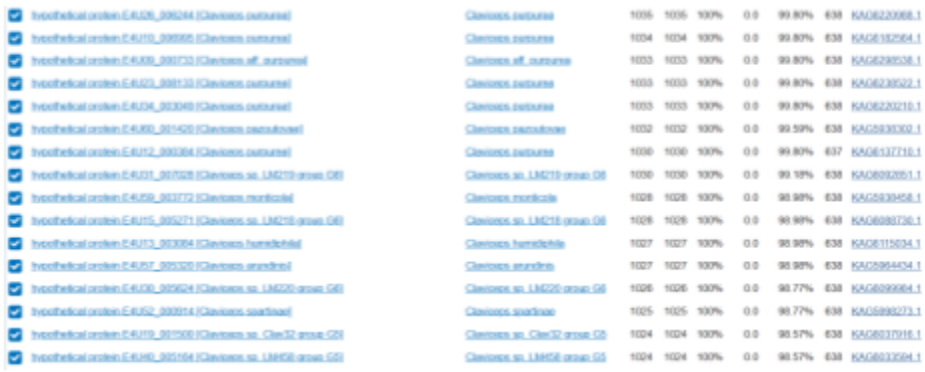


**Figure 1 BLAST Homology Algorithms.** The search query provides results for sequences homologous to CpCdc14. Top hits were ranked by alignment scores.

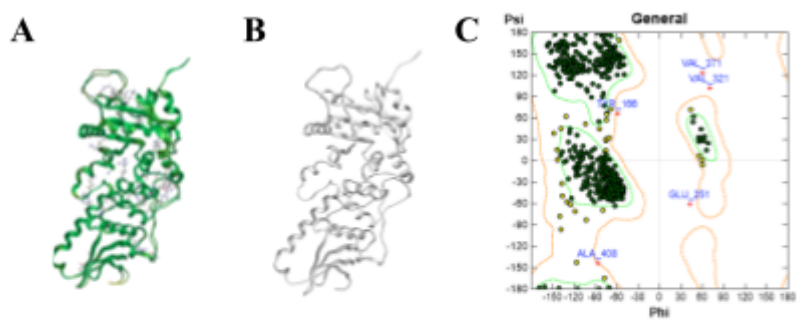



**Figure 2 Homology Model Overlay with Cdc14 Protein from Claviceps Purpurea.** Using the Molecular Operating Environment (MOE) software, a Cdc14 ortholog from a closely similar template species, tyrosine protein phosphatase Cdc14 (6G86.B), was superimposed onto the Claviceps Purpurea CDC14 analog. Regions were colored based on RMSD, with the green sections being areas of high similarity between the two structures, warmer, red colors as areas of differences, and white as areas of absences. **(A)** Homology model with aligned and superimposed tyrosine protein phosphatase template. **(B)** Homology model without aligned and superimposed tyrosine protein phosphatase template. **(C)** Ramachandran plot to demonstrate the distribution of Φ and Ψ dihedral angles in the *CpCdc14* protein structure.

Prior to initial laboratory experiments, programming was run to visualize the CpCdc14 amino acid model. Through an initial BLAST sequence comparison (Figure 1), CpCdc14 shares significant sequence homology with PTPases, as it had the lowest non-zero HMMER e-value ($7.8 \times 10^{53}$), a reliable means of statistical observation where the lowest EMMER e-value means a closer hit to the protein. Then, from a subsequent homology model generated from Molecular Operating Environment (MOE) software (Figure 2A), the *CpCdc14* ortholog was determined to have a percent identity of 66.8% and percent similarity of 83.70%. Because of this relatively

high comparative similarity, the molecular modeling algorithm was able to generate a structure that more closely resembled the original tyrosine-protein phosphatase ortholog in yeast. By tracking the amino acid residues, three main regions of substitutions to note were: Region 1: Asp72, Pro73, Glu74; Region 2: Thr202, Lys203, Ser204 (extreme); Region 3: His332, Gln333, Asn334.

**Figure 3 pET15b Recombinant Plasmid Diagram.** pET15b recombinant plasmid shows CpCdc14's DNA coding sequence inserted into the plasmid between cleavage sites NdeI and BamHI. SnapGene was used to display and edit this vector.

*Expression and Purification of CpCdc14*

After analyzing CpCdc14 using *in silico* methods, laboratory expression of the protein began with bacterial transformation. The plasmid recombinant vector containing both the CpCdc14 coding gene and the ampicillin resistant gene was successfully indoctrinated into the E. Coli, evident by the hundreds of bacterial colonies that were able to grow within the 100 µg/mL culture environment. After adding the L-Arabinose to begin the protein synthesizing process and purifying the protein, nineteen aliquots of 250 µL lysed protein solutions were obtained. The spectrophotometer experiment (Appendix B) coupled with the nanodrop spectrophotometer verified the protein concentration to be 11.14 µM. We were able to get good expressions here with a relatively high amount of purity. Without this, further vitro experiments would not have been able to run.

*SDS-PAGE Analysis of Purified Claviceps purpurea protein*

When comparing uninduced versus induced cells in Lanes 2 and 3, it is evident that there is a greater expression of *CpCdc14* in Lane 3 between 45 and 66.2 kDa (Figure 4). The darker, wider band is present in Lane 3, which suggests that the protein was successfully expressed. Furthermore, the *CpCdc14* protein is demonstrated to be enhanced when comparing Lanes 7 and 8, as Lane 7 contains 3 additional µg of purified *CpCdc14* protein when compared to Lane 8. From the Expasy Online Tool, our predicted molecular weight is about 5.46 kD, which lines up with the dark band present between 45 and 66.2 kDa in that lane.

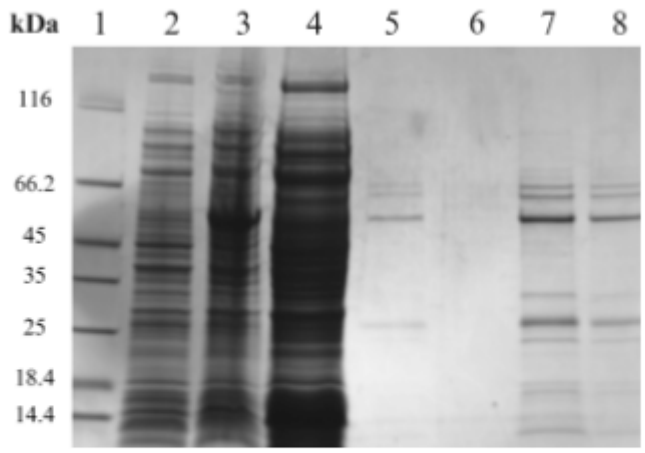



**Figure 4 SDS-PAGE of purified *Claviceps purpurea* protein stained with Coomassie brilliant blue.** Lane 1 contains protein molecular weight (MW) markers at the following values (in kDa): 116, 66.2, 45, 35, 25, 18.4, and 14.4. Lane 2 contains uninduced cells. Lane 3 contains induced cells. Lane 4 contains a sample of the second flow-through of Ni-NTA affinity chromatography. Lane 5 contains 250 mM imidazole elution 1. Lane 6 contains 250 mM imidazole elution 2. Lane 7 contains 5 µg of purified *CpCdc14* protein. Lane 8 contains 2 µg of purified *CpCdc14* protein.

*pNPP Assay +/- Sodium Tungstate*

After analyzing the DNA sequence using BLAST, the two domain hits with the lowest E values include Cdc14_C (E value = 9.98 x 10-84) and Cdc14_N (E value = 8.12 x 10-54). Thus, we hypothesize that the biochemical function of our target protein includes cell cycle regulation and cell division due to the biochemical function of CDC14 family proteins. To verify this, a pNPP Assay with sodium tungstate ($Na_2WO_4$), a general phosphatase inhibitor, was carried out to verify the identity of the purified protein. The concentration of the pNP product that formed from the added pNPP substrate for enzymes with and without sodium tungstate are 32.96 ± 0.13 and 0.28 ± 0.13 µM respectively. These results showed a drastic 99.15% inhibition (Figure 5), strongly suggesting that the protein is correctly hypothesized to be from the phosphatase family and a protein tyrosine phosphatase (PTP).

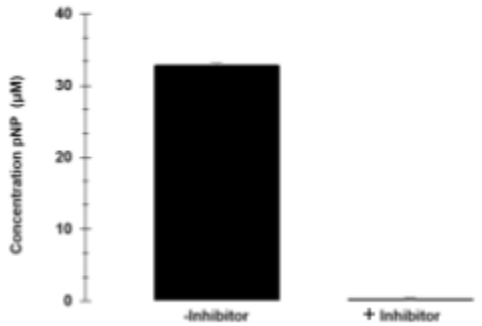



**Figure 5 pNPP Assay +/- Sodium Tungstate.** The three trials of our purified protein in solution with and without Sodium Tungstate inhibitor was graphed, displaying the average final concentration product of pNP as 32.96 and 0.28 µM respectively.

*Michaelis-Menten Plot and Steady-State Kinetic Parameters*

The literature kinetic parameters of Cdc14 phosphatase catalysis in *Saccharomyces cerevisiae* (ScCdc14) through a p-nitrophenyl phosphate (pNPP) substrate were provided under steady-state conditions: $k_{cat}$ (s-1) = 1.12 ± 0.08 , $K_M$ (mM) = 23.0 ± 1.0, and $k_{cat}$ • $K_M$-1 ($s^{-1}$ • $mM^{-1}$) = 50 ± 4.0 After running our steady state kinetic assays, the average calculated $K_M$ value was 35.59 ± 3.20, which is within the acceptable range of existing $K_M$ values and $V_{max}$ was calculated to be 2.16 ± 0.33 (Figure 6).

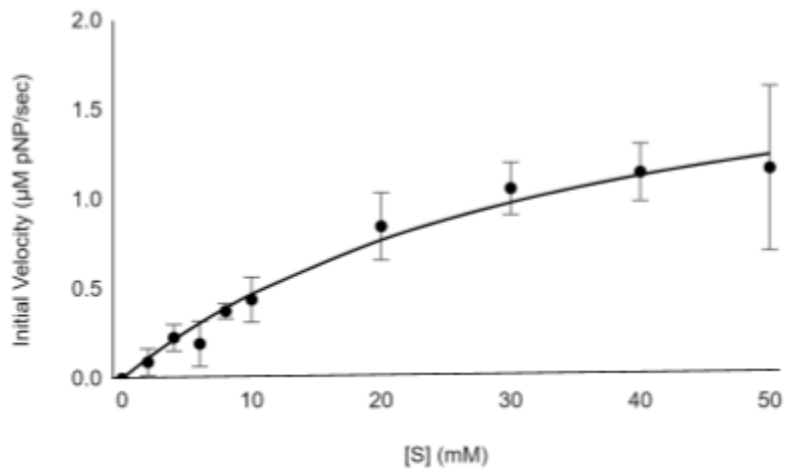



**Figure 6 Michaelis-Menten Plot and Steady-State Kinetic Parameters of CDC14 Protein with pNPP Substrate.** Concentrations of 2, 4, 6, 8, 10, 20, 30, 40, 50, 60, 70, and 80 mM of pNPP were added to enzyme well solutions, and absorbance was calculated after 20 minutes of enzyme activity. Error bars were added to represent standard deviation for the three trials as shown in the figure.

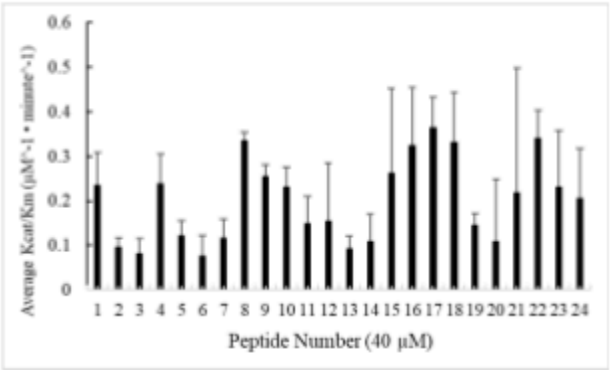
*Substrate Specificity of CpCdc14*

Our goal was to find the substrate with the most effective binding and rate of catalysis, which can then be used as a model for the future potential inhibitor design. Biomol Green reagent dye was added to every well to react with free phosphates and left to incubate. The

absorbance (concentration of free phosphates) of each well was measured at 605 nm using the BioTek plate reader. The previously created Biomol green standard curve (Appendix C) was used to estimate the concentration based on the absorbance.

This number was then used to calculate the $k_{cat}/K_M$ for each of the 24 inhibitors. The $k_{cat}$ was calculated with the equation: $k_{cat} = V_{max} \cdot [E]_T^{-1}$ and a value of 35.60 was used for $K_M$ . The most effective substrates were found to be peptides #17, #22, #18, and #8, as they had the highest calculated $k_{cat}/K_M$ values, 0.3657045177, 0.3424324121, 0.3324586525, and 0.335783239, respectively (Figure 7A). All of the most effective substrates share a phosphorylated serine group, a proline in the +1 position, and a lysine in the +3 position, which was seen in our own data and across groups testing other Cdc14 orthologs. (Figure 7B) The most unsuccessful substrate wells with the lowest $k_{cat}/K_M$ values, like peptides #6 and #3, lacked the previously stated characteristics in their sequences. Previously conducted research [12] of other Cdc14 orthologs identified similar required traits for an effective substrate.

**A**




**B**

| Substrate Number: | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Sequence: | HT{pS}PIKSIG | HT{pT}PIKSIG | HT{pY}PIKSIG | HT{pS}AIKSIG | HT{pS}GIKSIG | HT{pS}GIASIG | HT{pS}GIKRIG | HT{pS}PKKSIG |
| Substrate Number: | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| Sequence: | HT{pS}PRKSIG | HT{pS}PEKSIG | HT{pS}PPKSIG | HT{pS}PIASIG | HT{pS}PIRSIG | HT{pS}PILSIG | HT{pS}PIKKIG | HT{pS}PIKRIG |
| Substrate Number: | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |
| Sequence: | HT{pS}PIKSKG | HT{pS}PIKSRG | HT{pS}PIKASIG | HT{pS}PIAKIG | HT{pS}PIASKG | HT{pS}PKKRIG | RT{pS}PIKSIG | HD{pS}PIKSIG |

**Figure 7 Substrate Specificity of *CpCdc14*. (A)** Bar graph describing enzyme kinetics of each peptide number in relation to *CpCdc14*. **(B)** Table listing the acid sequences of each phosphopeptide sequence tested. {pS} indicates a phosphoserine peptide. {pT} indicates a phosphotyrosine peptide. {pY} indicates a phosphotyrosine peptide.

*In silico Inhibition of CpCdc14*

After docking the 11 inhibitor compounds in MOE in order to find the most effective inhibitor with the best binding affinity, the highest S-values, indicating substrate binding affinity, were reported for each inhibitor. For A9, S = -5.5044; D7, S = -5.0295; D3, S = -5.6096; E1, S = -5.6902; G5, S = -5.1625; G6, no binding affinity recorded; G7, S = -5.3395; H7, S = -4.7399;

I1, S = -5.8362; I2, no binding affinity recorded; I9, S = -4.7816. Based on *in silico* modeling, the three inhibitors with the greatest binding affinity include: I1, E1, and D3.

*In vitro Inhibition of CpCdc14*

The purpose of this experiment was to find the most effective inhibitor for CpCdc14. Positive and negative controls and blanks were run in order to calculate the Z-factor. The calculated Z-factor was 0.928, which is greater than 0.5, which is the baseline for statistically significant data. This meant that the chosen concentrations and solution amounts were adequate enough to continue running assays for the ten inhibitors.

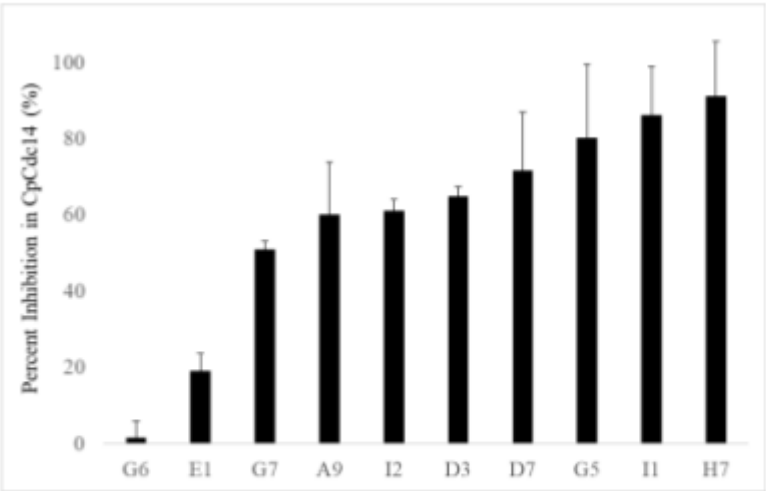



**Figure 8 In vitro Inhibition Screening.** The bar chart shows the percent inhibition for each of the ten tested inhibitors. This data was used to determine which inhibitors are the most effective and should be further tested.

Based on the results of the ten inhibitor assay (Figure 8) and the predictions on the Molecular Operating Environment (MOE), IC50 curves were created for the top two most effective inhibitors: I1 and G5 (H7 was not ran further because there was not an adequate amount of inhibitor to run enough trials). I1, a cyclic phosphoprotein, was chosen as first with average percent inhibition being 86.21%. G5 was chosen as second with average percent inhibition being 80.17%. The calculated IC50 was 0.219 for I1 and 0.119 for G5 (Figure 9A and 9B). This low value meant a high inhibition.

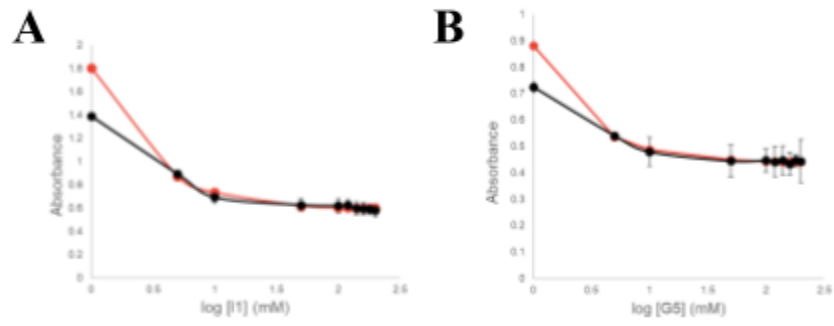



**Figure 9 log [Inhibitor] vs Absorbance.** After determining inhibitors I1 and G5 as the most effective via in vitro experiments, inhibition assays were run and graphed to determine IC50 values. The black curve in both graphs represents the experimentally obtained results, while the red curve represents the theoretical values as modeled by a sinusoidal equation.

*In silico Optimization of Aurintricarboxylic acid (Inhibitor I1)*

Based on *in silico* and *in vitro* analyses, I1 was determined as a top inhibitor to further optimize. The final binding affinity prediction (Figure 11A), represented by S ($k_{cal} \cdot mol^{-1}$), was optimized from -7.56 to -10.70 (-3.14). A lower S-value indicates a greater binding affinity. After analyzing electrostatic maps and finding a hydrophobic region represented by white, a methyl ($-CH_3$) group was added. The S-value decreased to -7.91. Next, a favorable H-bond donor (O-H) was added to the I1 molecule. The S-value decreased to -8.13. Based on the Fragment capabilities within MOE, a 5-imidazole and cyclopropyl group were added, in which the S-values decreased to -9.15 and -9.99, respectively. To continue, based on favorable regions, a methanol and acetyl group were added, in which the S-values decreased to -10.40 and then -10.70. Ligand interaction maps (Figures 10A and 10D) were also utilized to determine areas of ligand exposure and preferable interactions, such as being a sidechain acceptor region.

A modified Lipinski's rule of five was followed, as: log(P) or octanol-water partition coefficient, remained below 5.00 (final = 4.94), there were no more than 5 H-donors (final = 4), there were no more than 10 H-acceptors (final = 10), and the molecular weight remained below 750 Da (final = 639.57 Da). The overall structural change from initial to optimized I1 structure is described in *B* and *C* of Figure 12. Moreover, the initial and optimized inhibitor molecules, in the context of the *CpCdc14* pocket, are seen in Figures 10A and 10C.

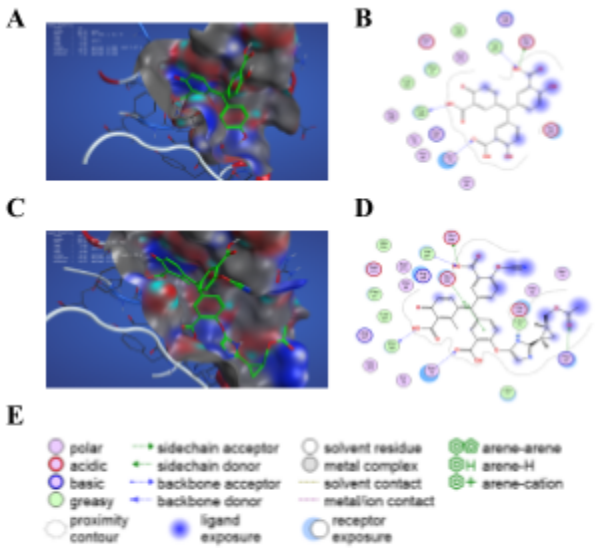



**Figure 10 In silico optimization of aurintricarboxylic acid (Inhibitor I1). (A)** Initial I1 inhibitor structure (green) with receptor surface and pocket atoms displayed. **(B)** Two-dimensional ligand interactions map between *CpCdc14* homology model pocket and initial I1 inhibitor. **(C)** Optimized I1 inhibitor structure (green) with receptor surface and pocket atoms displayed. **(D)** Two-dimensional ligand interactions map between *CpCdc14* homology model pocket and optimized I1 inhibitor. **(E)** Legend for (B) and (D).

A

| Additions to I1 Structure | S (kcal • mol-1) | ΔS (kcal • mol-1) | log P | Δ log P | MW (g • mol-1) | Δ MW (Da) |
|---|---|---|---|---|---|---|
| Initial | -7.56 | 0.00 | 2.97 | 0.00 | 422.34 | 0.00 |
| Methyl group | -7.91 | -0.35 | 3.46 | 0.49 | 436.37 | 14.03 |
| H-bond donor (O-H) | -8.13 | -0.22 | 3.46 | 0.00 | 436.37 | 0.00 |
| 5-imidazole | -9.15 | -1.02 | 3.78 | 0.32 | 502.43 | 66.06 |
| Cyclopropyl group | -9.99 | -0.84 | 4.93 | 1.15 | 567.51 | 65.08 |
| Methanol | -10.40 | -0.41 | 4.56 | -0.37 | 597.54 | 30.03 |
| Acetyl group (final) | -10.70 | -0.30 | 4.94 | 0.38 | 639.57 | 42.03 |

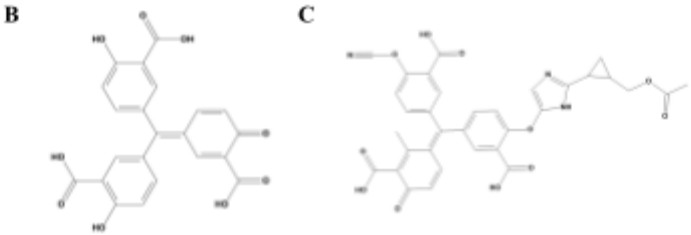



**Figure 11 Documented modifications to initial Aurintricarboxylic acid (I1) inhibitor structure. (A)** Additions to I1 structure in a tabular format. **(B)** Initial I1 skeletal structure. **(C)** Optimized I1 skeletal structure.

## Discussion

In this study, our overall goal was to provide support for the potential of fungal Cdc14 enzymes as new pesticide targets, through the characterization of a *Claviceps purpurea* Cdc14 ortholog (*CpCdc14*). We initially hypothesized that CpCdc14 would display characteristics similar to other protein-tyrosine phosphatases (PTPs), and that it would serve as a promising target for fungicide development. Through a combination of *in silico* analyses and *in vitro* assays, we have made significant progress in understanding the structure, specificity, and inhibition potential of *CpCdc14*. Our results from sequence and homology modeling suggested that CpCdc14 is, in fact, a tyrosine-protein phosphatase ortholog, sharing significant similarity (66.8% identity and 83.7% similarity) with a template protein tyrosine phosphatase from budding yeast (*Saccharomyces cerevisiae; ScCdc14*). Molecular Operating Environment (MOE), allowed us to generate a structure that closely resembled the yeast ortholog, providing valuable insights into the potential function and interactions of *CpCdc14*.

SDS-PAGE analysis of the purified CpCdc14 protein is an essential step in evaluating the success of the purification process and assessing protein purity. The gel clearly demonstrated that the *CpCdc14* protein was enriched after $Ni^{2+}$-NTA affinity chromatography. Comparing uninduced and induced cell lysates, we observed a distinct band between 45 and 66.2 kDa, indicating the presence of CpCdc14 in the induced cells and its purification through affinity chromatography. This outcome supports the idea that CpCdc14 is well-expressed in *E. coli* (pET-15b plasmid) and can be produced at a relatively high level. However, it is important to acknowledge that some faint bands were also present, indicating possible impurities or

intracellular fragments. While the overall purity appears satisfactory, further optimization of purification steps may be needed in later research requiring further purified *CpCdc14*.

The pNPP assay was crucial in verifying the identity of *CpCdc14* as a PTP. The significant 99.15% inhibition of the enzyme's activity in the presence of sodium tungstate strongly supports the hypothesis that *CpCdc14* is indeed a phosphatase–supporting the BLAST DNA sequence analysis. This finding is consistent with the *in silico* prediction using MOE and the homology to known PTPs. Furthermore, the conducted Michaelis-Menten kinetics assay provided insight into the enzymatic properties of *CpCdc14*. The calculated $K_M$ value (35.59 ± 3.2) for *CpCdc14*'s pNPP substrate is consistent with existing $K_M$ values for Cdc14 enzymes (e.g. *ScCdc14*); although slight variations in $K_M$ from literature review was present, this may arise from species-specific differences or experimental variations. Such results suggest that CpCdc14 has a comparable affinity for its substrate as other Cdc14 orthologs, reinforcing its functional conservation.

Substrate specificity analysis revealed four peptides (17, 22, 18, and 8) with high specificity to *CpCdc14*, exhibiting a phosphorylated serine group, a proline in the +1 position, and a lysine in the +3 position. These results are consistent with previous studies of Cdc14 orthologs (12), which have also identified similar substrate requirements. This provides evidence substrate specificity conservation within the Cdc14 family, reinforcing *CpCdc14*'s potential as a potential broad spectrum target.

The high-throughput in vitro inhibition assay allowed us to screen a set of potential inhibitors and identify the top three compounds (I1, E1, and D3) with the greatest binding affinity to CpCdc14. The IC50 curves for these inhibitors provide a quantitative measure of their inhibitory potential, enabling us to prioritize and focus on the most promising compounds. Our *in vitro* inhibition assays with 11 potential inhibitors filtered from the high throughput screening shed light on three top-performing compounds (I1, G5, and D7) with high binding affinity to CpCdc14. However, upon testing the inhibitors in the laboratory, the top scoring inhibitors were I1 and G5. This suggests that there was a limit on MOEs predictions that were not fully captured in the laboratory experiments.

As aurintricarboxylic acid (I1) had the most favorable *in silico* results in *CpCdc14* inhibitor, steps were taken to optimize its binding affinity with the *CpCdc14* active site. Taking electrostatic mapping, protein-ligand interaction diagrams, and a structural analysis of I1 into account, the following favorable modifications were made (in sequential order): adding a methyl group, H-bond donor (O-H), 5-imidazole, cyclopropyl group, methanol, and acetyl group. The predicted binding affinity, represented by *S* ($k_{cal} \cdot mol^{-1}$), favorably decreased by 3.14, as lower values indicate enhanced affinity. Moreover, the optimization of I1 followed a modified Lipinski's rule of five, as log(P) or octanol-water partition coefficient, remained below 5.00 (final = 4.94), there were no more than 5 H-donors (final = 4), there were no more than 10 H-acceptors (final = 10), and the molecular weight remained below 750 Da (final = 639.57 Da). Future research can investigate methods of increased binding affinity by exploring other functional groups, organic molecules, such as pyrrolidine, and the potential use of amino acid residues. However, some current modifications may need to be reversed to allow for the octanol-water partition coefficient to remain below 5.00.

Future work should focus on developing and optimizing *CpCdc14*-specific inhibitors, while testing their efficacy against similar pathogenic fungal targets, especially ascomycetes. As Cdc14 phosphatases are not present in many other eukaryotes, *CpCdc14* has significant potential as a fungicidal target. Future research can also examine large-scale applications of a novel fungicide to combat global food insecurity and undernourishment.

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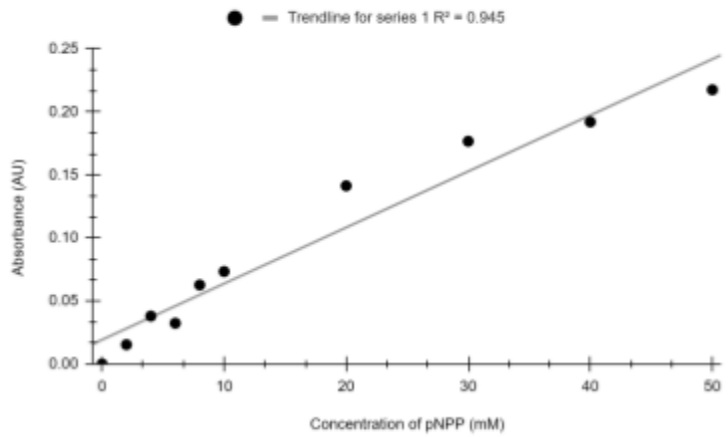
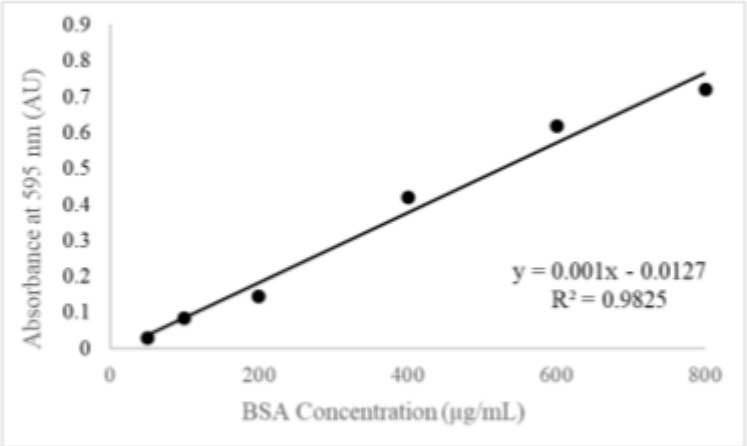
**Appendix**

*Appendix A:* pNPP Standard Curve



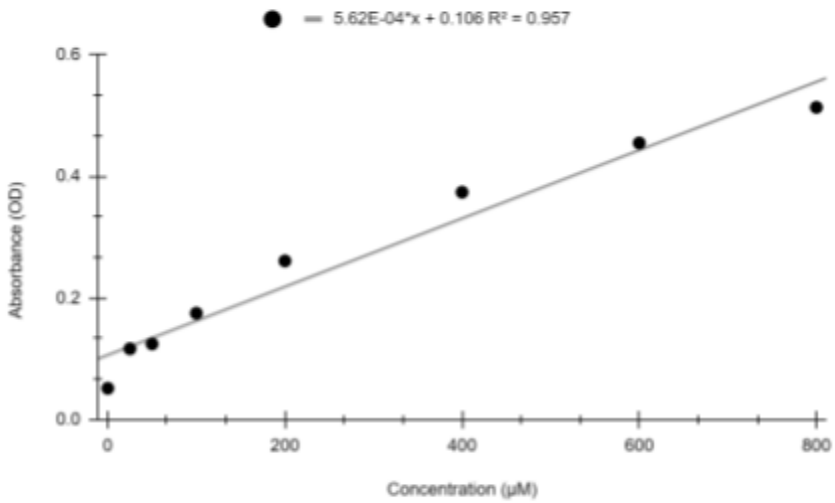

**Appendix A pNPP Standard Curve.** The pNPP standard curve plots concentration of pNPP against the measured absorbances.

*Appendix B:* BSA Standard Curve




**Appendix B Bradford Assay.** The Bradford Assay standard curve plots absorbance versus a variety of diluted bovine serum albumin (BSA) stocks measured at 450 nm.

*Appendix C:* Biomol Green Standard Curve




**Appendix C Biomol Standard Curve** The curve plots absorbance values for diluted solutions of sodium phosphate measured at 640 nm. The graph shows the linear range for the specificity assay, allowing for the conversion of absorbance measurements to product formation.

*Appendix D:* Genetic Coding Sequence

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>Claviceps purpurea gene coding sequence
ATGGCGCCTTCCAAGAACGCGTCCAACATGGGCCAAATTATTGAGTATATTCCGGATCGCTTGTATCTCGGCGCCTACG
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*Appendix E:* Amino Acid Sequence

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>Claviceps purpurea amino acid sequence
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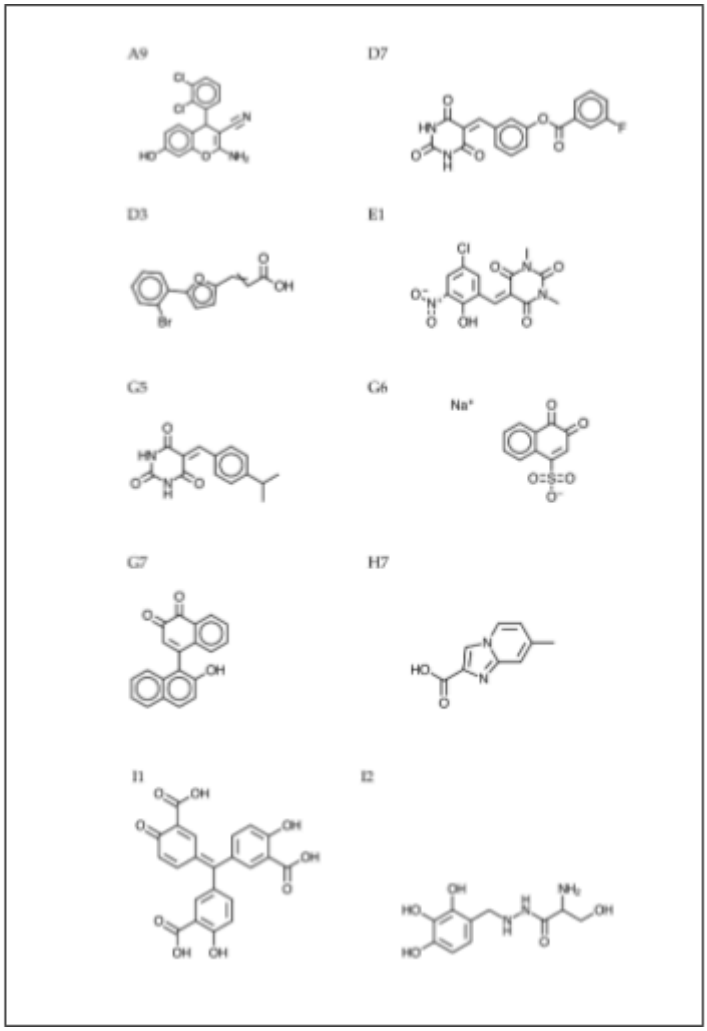
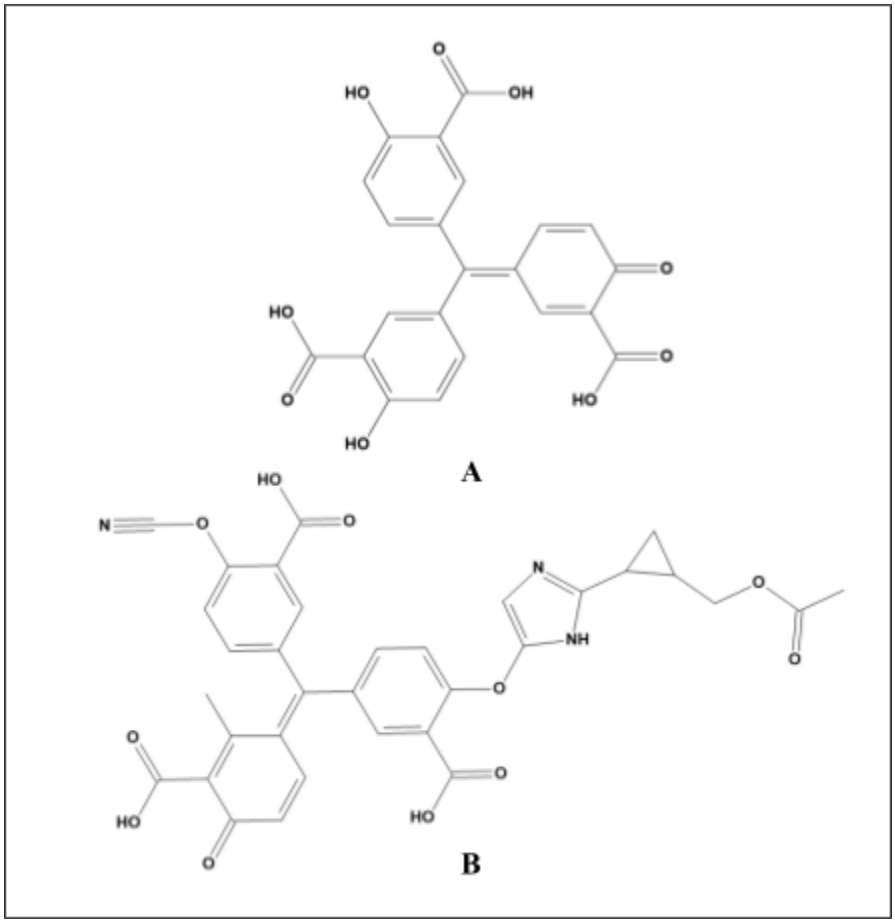
*Appendix F:* Structure of Inhibitors



*Appendix G:* Initial vs. Optimized *I1* Structures